

CS 401 — Advanced Algorithms

Course Syllabus & Study Guide — Fall 2026

Section 1: Course Overview

CS 401 covers the design and analysis of algorithms at an advanced undergraduate level. Topics include sorting networks, amortized analysis, randomised algorithms, NP-completeness, approximation algorithms, and parallel computing models. The course assumes familiarity with data structures, discrete mathematics, and basic probability theory.

Section 2: Instructor & Schedule

Field	Details
Instructor	Dr. Amina Siddiqui
Email	a.siddiqui@university.edu
Office Hours	Tuesday & Thursday 3:00 PM – 5:00 PM, Room 214B
Lecture Days	Monday, Wednesday, Friday — 10:00 AM to 11:15 AM
Classroom	Engineering Block, Hall C (Room EC-302)
Term Dates	1 September 2026 – 18 December 2026

Section 3: Grading Policy

Final grades are computed from the following components. There are no make-up exams without documented medical or personal emergency approved by the department.

Component	Weight	Notes
Assignments (6 total)	25%	Due every two weeks, submitted on Moodle
Midterm Examination	25%	Week 8, in-class, closed-book
Final Project	20%	Group of 2–3, report + 15-min presentation
Final Examination	25%	Cumulative, open one page of notes
Participation	5%	In-class quizzes and discussion

Section 4: Weekly Topics

Weeks 1–2: Review: Recurrences, Master Theorem, and Divide & Conquer

Weeks 3–4: Greedy Algorithms: Activity Selection, Huffman Codes, Matroid Theory

Weeks 5–6: Dynamic Programming: Chain Matrix Multiplication, Longest Common Subsequence, Knapsack

Weeks 7–8: Graph Algorithms: Bellman-Ford, Johnson's Algorithm, Maximum Flow (Ford-Fulkerson, Dinic)

Week 9: MIDTERM EXAMINATION — covers Weeks 1–8

Weeks 10–11: Randomised Algorithms: Randomised QuickSort, Karger's Min-Cut, Monte Carlo vs Las Vegas

Week 12: Amortised Analysis: Aggregate, Accounting, and Potential Methods; Splay Trees

Week 13: NP-Completeness: Cook-Levin Theorem, Common NP-Complete Problems and Reductions

Week 14: Approximation Algorithms: Vertex Cover, TSP Approximation, PTAS

Week 15: Parallel Algorithms: PRAM Model, Parallel Sorting, Work-Depth Analysis

Week 16: Final Project Presentations & Review

Section 5: Late Submission & Academic Integrity Policy

Assignments submitted up to 24 hours late receive a 15% penalty. Submissions between 24 and 48 hours late receive a 40% penalty. No submissions are accepted more than 48 hours after the deadline without prior approval. All work must be your own. Collaboration is encouraged for understanding concepts, but submitted code and write-ups must be written independently. Any form of plagiarism will result in a zero on the assignment and may be referred to the Academic Integrity Office.

Section 6: Required Textbooks

- Cormen, Leiserson, Rivest, Stein — *Introduction to Algorithms*, 4th Edition (MIT Press).
Primary text.
- Kleinberg & Tardos — *Algorithm Design*, 1st Edition (Pearson). Supplementary.
- Mitzenmacher & Upfal — *Probability and Computing*, 2nd Edition (Cambridge). For randomised algorithms unit.

This syllabus is subject to change. Any changes will be announced in class and posted on Moodle.